

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S) & LIIT **Paper -2(Adv-2024-P2-Model)** **Date: 21-12-2025**

Time: 02.00Pm to 05.00Pm **GTA-08** **Max. Marks: 180**

21-12-2025_Sr. XL-500, Sr.S60(N&S) & LIIT_Jee-Adv(2024-P2)_GTA-08_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCED-2024-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. Let $f(x,y)$ be a locus of a point $P(x,y)$ satisfying $\alpha(2x-y+1)+\beta(3x-y)+\gamma(2x+y-5)=0 \forall \alpha,\beta,\gamma \in R$. The least distance between the curve $f(x,y)$ and straight line $3x-4y+19=0$ is.

A) 3 **B)** 2 **C)** $\frac{7}{5}$ **D)** $\frac{26}{5}$

2. If $\frac{\cos x+\cos y+\cos z}{\cos(x+y+z)}=2$ and $\frac{\sin x+\sin y+\sin z}{\sin(x+y+z)}=2$, then the value of $\cos(x+y)+\cos(y+z)+\cos(z+x)$ is equal to : $(\text{where } x,y,z \in R)$

A) 3 **B)** 1 **C)** 2 **D)** – 1

3. Let $f(x)$ and $g(x)$ be continuous, positive function such that $f(-x)=g(x)-1, f(x)=\frac{g(x)}{g(-x)}$ *and* $\int_{-20}^{20} f(x)dx=2020$, then the value of $\int_{-20}^{20} \frac{f(x)}{g(x)}dx$ is

A) 1010 **B)** 1050 **C)** 2020 **D)** 2050

4. Let $g(x)=6x^2-18x+8, f_1(x)=|g(x)|, f_2(x)=|f_1(x)-P_1|, f_3(x)=|f_2(x)-P_2|$ and if $P_1=7$, then the range of P_2 such that $f_3(x)$ has exactly 10 points of non-differentiability is

A) $\left(\frac{3}{2},7\right)$ **B)** $\left[\frac{5}{2},8\right]$ **C)** $[2,9]$ **D)** $(1,8)$

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
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Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. Let $f:(1-\infty)\to R$ be a differentiable function such that

$$\int_2^{150}(x-1)\ln(x-1)\left(2f(x)-(x-1)\ln(x-1)\right)dx=\int_2^{150}f^2(x)dx.$$ Then

A) Area bounded by the curve and x-axis is equal to $\frac{1}{2}$.

B) $f(x)$ is strictly decreasing in $\left(1,1+\frac{1}{e}\right)$.

C) Number of solutions of the equation $f(x)=2$ is 2.

D) $f(x)$ is monotonic in $\left(1+\frac{1}{e},\infty\right)$.

6. Let $f(x)=ax^2+bx+c, a,b,x\in R, a\neq 0$.

If $f(2018)-f(-2014)=\dfrac{\sin^2\left(\left[\tan^{-1}x^2\right]\pi\right)+3\tan^2\left(\left[\cot^{-1}x^2\right]\pi\right)}{3\sin^2 x+\cos^2 x}\forall x\in R$ and

$f(2018)+f(-2014)=2(2016)^2+12$, then :

[Note : $[k]$ denotes greatest integer function less than or equal to k and $\text{sgn}(k)$ denotes signum function of k.]

A) maximum value of $f(x)$ is 6.

B) $f(1)+f(2)+f(3)$ is equal to 20.

C) minimum value of $f(f(f(x)))$ is 490.

D) number of solution(s) of the equation $f(x)=\text{sgn}(f(x))$ is 0.

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7. Let the set of complex numbers $(a_1,b_1),(a_2,b_2),(a_3,b_3)$..... denoting the points on the complex plane satisfying $(a_{n+1},b_{n+1})=(\sqrt{3}a_n-b_n,\sqrt{3}\,b_n+a_n)$ *for* $n=1,2,3$.....

Suppose $(a_{100},b_{100})=(2,4)$, then

A) $(a_1+b_1)=\frac{1}{2^{96}}$ **B)** $2(a_1+b_1)=\frac{1}{2^{97}}$

C) $(a_1+b_1)=\frac{1}{2^{98}}$ **D)** $\frac{(a_1+b_1)}{2}=\frac{1}{2^{99}}$

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

8. Let E and M be 3×3 matrices satisfying the system of equations

$EM^T=(EM)^T=20I$

and $(E+M)^T=17(E-M)^T$

Where I denotes identify matrix of order 3.

If $E^2+M^2=\frac{a}{b}I$ (where a and b are co-prime), then find the value of $(a+b)$.

9. Let $a_1,a_2,...,a_n$ be the roots of the polynomial $\sum_{k=1}^{n}kx^k=0$. If $\sum_{k=1}^{n}\frac{1}{(1-a_k)^2}=-13$.

Find n.

10. Let S_k be the area bounded by the curve $y=x^2(1-x)^k$ and the lines $x=0, y=0$ and $x=1$. If $\lim_{x\to\infty}\sum_{k=1}^{n}S_k$ is equal to $\frac{p}{q}$, where p and q are co-prime positive integers, find $(p+q)$.

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11. Two parallel planes are given by, $x+y+z=1$ and $x+y+z=\frac{9}{2}$. A third plane that intersects them is given by $2x-5y+z=-5$, resulting in two parallel lines of intersection. If the distance 'd' between these two parallel lines can be expressed as $d=\sqrt{\frac{a}{b}}$, where a and b are co-prime positive integers, then find the value of $[d]$.

[Note : where $[k]$ denotes greatest integer function less than or equal to k.]

12. If $f(x)=(x-a)(x-b)$ for $a,b\in R$, then find the minimum number of roots of equation $\pi(f'(x))^2\cos(\pi(f(x)))+\sin(\pi(f(x)))f''(x)=0$ in $[\alpha,\beta]$ where $f(\alpha)=3=f(\beta)$ and $\alpha<a<b<\beta$.

13. In ΔABC, let $|a-5|+(7\cos B-3)^2+\log_3(1+|c-7|)=0$. A circle drawn with the altitude AD as diameter which meets the sides AB and AC at E and F respectively. If $(BE)(CF)=\frac{p}{q\sqrt{11}}$ where $p,q\in N$, then find the least value of $(p+q)$.

SECTION-IV
(PARAGRAPH QUESTIONS)

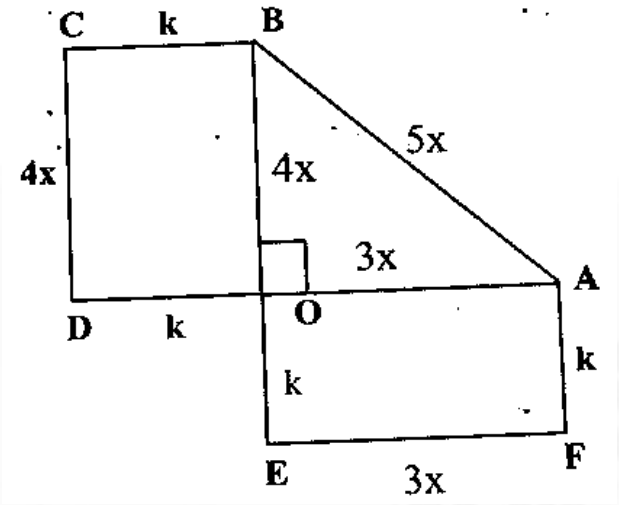
- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Figure shown two rectangles and a right angled triangle attached together and it's dimensions are given.

Perimeter of the 7-sided polygon OEFABCD is P units and area of the 7-sided polygon OEFABCD is A sq. units

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14. If P = 30 units and 'A' is maximum possible value, then $x =$ _____

15. If A = 13 sq. units and 'P' is maximum, then $x^2 =$ _____

PARAGRAPH-II

A quadratic polynomial $f(x)$ with positive leading coefficient such that $g(x) = f(\ln x) \forall x > 0$. Also the curve $y = g(x)$ satisfy following condition.

a) There is exactly one value for a positive number p such that $(p, g(p))$ is its extremum point and $\left(p^2, g\left(p^2\right)\right)$ is its inflection point.

b) Exactly one tangent line can be drawn from the point $(0,0)$ to the curve $y = g(x)$.

16. The value of $\dfrac{f(10)}{10f(2)}$ equals

17. The value of definite integral of $\displaystyle\int_1^2 \frac{f(0)}{f(x)}dx$ equals (Take $\pi = 3.14$)

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PHYSICS **Max Marks: 60**

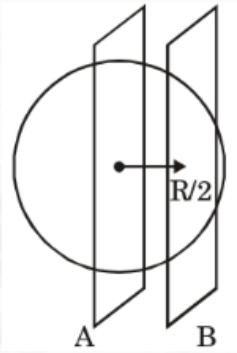
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

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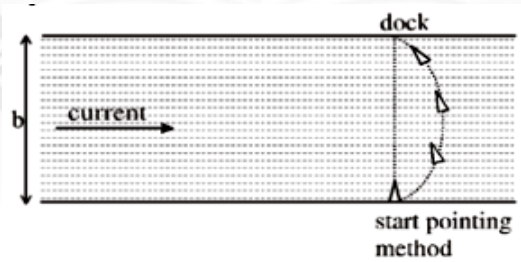
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

18. From a sphere of electric conductivity K two planes cut a piece such that first plane passes through the centre of sphere and second parallel to first one at distance $\frac{R}{2}$ from centre then resistance between A and B.



A) $\frac{1}{\pi KR}$ **B)** $\frac{\sqrt{3}}{\pi KR}$ **C)** $\frac{\ln 3}{2\pi KR}$ **D)** None

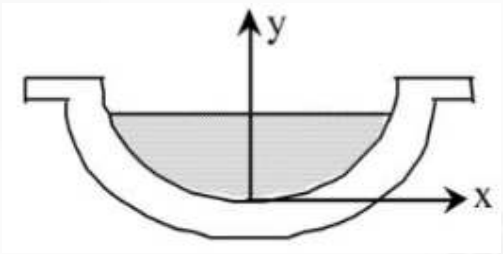
19. Suppose you are anchored near the shore of a channel in which there is steady current, and you are going to run your (motor) boat at constant throttle to a dock directly across the channel on the opposite shore. The way in which one might steer the boat to the dock is pointing method. In pointing method keep the nose of the boat pointed directly at the dock. Velocity of boat with respect to water is 'v' and velocity of water is 'u'. Width of water channel is 'b'. The time taken by boat to reach the dock is



A) $\frac{bv}{v^2-u^2}$ **B)** $\frac{b}{\sqrt{v^2-u^2}}$ **C)** $\frac{bu}{v^2}$ **D)** $\frac{bv}{u^2}$

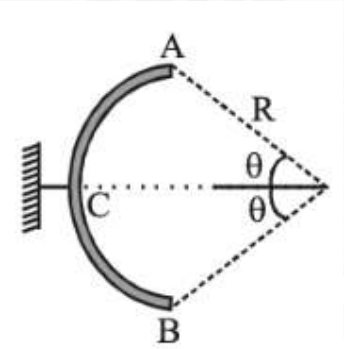
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20. A small hole is made at the bottom of a symmetrical jar as shown in figure. A liquid is filled into the jar upto a certain height. The rate of descension of liquid is independent of the level of the liquid in the jar. Then the surface of the jar is a surface of revolution of the curve (K is constant)



A) $y = kx^4$ **B)** $y = kx^2$ **C)** $y = kx^3$ **D)** $y = kx^5$

21. A uniform rod AB is bent in the shape of an arc of circle and hinged at point C. Determine the angular acceleration of the rod immediately after it is released from rest in a vertical plane.



A) $\frac{g}{R}\sin\theta$ **B)** $\frac{2g}{R}\sin\theta$ **C)** $\frac{g}{2R}$ **D)** $\frac{g}{2R}\left(\frac{5\sin\theta}{\theta}\right)$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

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22. Positronium is a system consisting of an electron and its antiparticle a positron, bound together by their electrostatic attraction. Assume that both particles revolve about their common centre of mass. Mark the CORRECT options remaining in the framework of Bohr hydrogen atom model. Use $k_e = \frac{1}{4\pi \in_0}, \hbar = \frac{h}{2\pi}$ and standard notations.

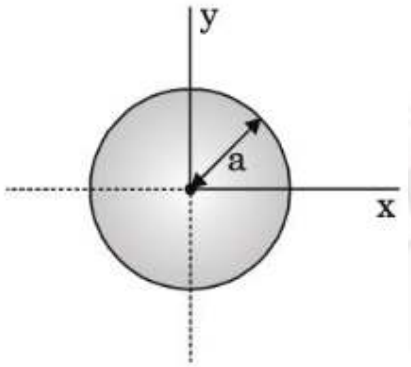
A) Radius of nth orbit is $r_n = \left(\frac{\hbar}{mk_e^2e^2}\right)n^2$

B) Speed in n^{th} orbit is $u_n = \left(\frac{k_e e^2}{2\hbar}\right)\frac{1}{n}$

C) Total energy of the bound state is $E_n = -\frac{m\left(k_e e^2\right)^2}{4\hbar^2} \times \frac{1}{n^2}$

D) The magnitude of the energy levels are exactly one-half of the corresponding value of Bohr's H - atom.

23. A circular ring of radius a and resistance R is placed (fixed) in x-y plane with its centre at origin as shown in the figure. A magnetic field $\vec{B} = (\alpha x)\hat{i} + (\beta y)\hat{j} + (\gamma t)\hat{k}$ is switched on at $t = 0$.



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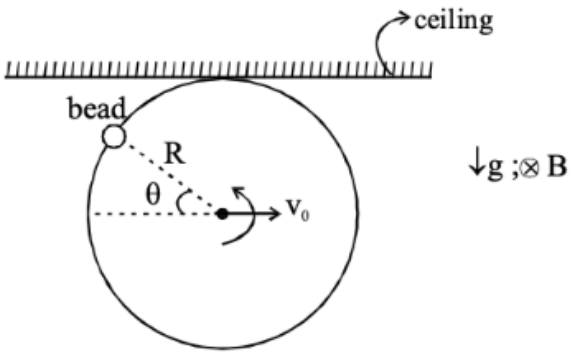
A) Current induced in ring is $\frac{\pi a^2 \gamma}{R}$

B) Current induced in ring is $\frac{\pi a^2 \gamma}{2R}$

C) The magnitude force acting on the ring is $\frac{\pi^2 a^4 (\alpha+\beta)\gamma}{2R}$

D) The magnitude force acting on the ring is $\frac{\pi^2 a^4 (\alpha+\beta)\gamma}{R}$

24. A insulating ring of radius R, is rolled uniformly along the ceiling of a room with constant velocity v_0. A charged bead of mass m, charge q is fixed on the ring. Entire system is moving in an uniform magnetic field as shown in the figure. The bead moves along with ring without any contact force from ring. $(\theta \neq 90°)$. Choose the correct statements



A) The centripetal force on bead from the frame of refrance of centre of mass of ring is qv_0B

B) The magnetic field $B = \frac{mg}{qv_0}$

C) The radius of the ring is $R = \frac{mv_0}{qB}$

D) The tangential acceleration of bead is $\frac{qv_0B\cos\theta}{m}$

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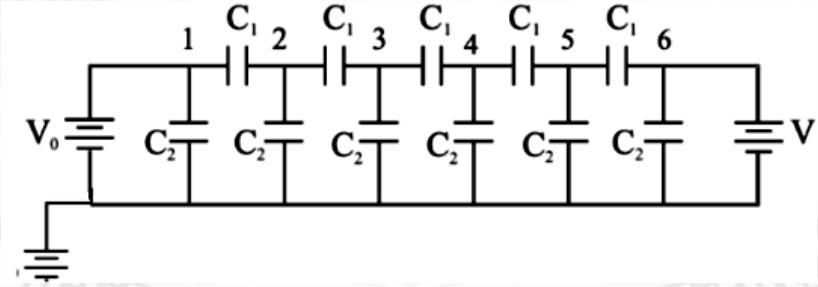
SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
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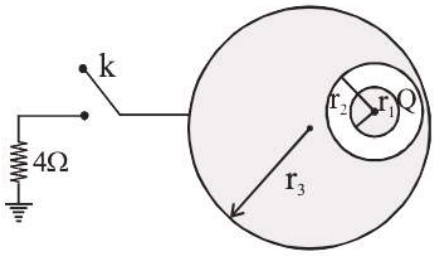
25. A small block carrying a charge $+Q$ is placed on a frictionless horizontal table and is attached to one end of a spring of force constant k. The other end of the spring is fixed to a support, and the spring is initially unstretched and horizontal. Another small block with charge $-q$ is brought in very slowly from a large distance toward the first block, moving along the line of the spring's axis. The extension of the spring at the instant the two blocks make contact is $\frac{a}{b}\left(\frac{Qq}{2\pi \in_0 k}\right)^{\frac{1}{a}}$. Determine the value of $(a+b)$.

26. In the circuit shown, ratio of capacitors is $\frac{C_2}{C_1}=\frac{4}{3}$ and terminal 'voltage of the battery connected at the left end is V_0. If the terminal voltage of the battery connected at the right most end is in such a way that the potentials of nodes 1, 2, 3, 4, 5 and 6 -are in a geometric progression. If $V=V_0x^y$, then the value of $|x|+|y|$ is: (Note: x & y are integers)



27. Charge $Q=2\mu C$ is distributed on a small conducting sphere $(\text{radius}\, r_1=0.1m)$ as shown inside a spherical cavity of radius $r_2=0.5\,m$ made inside a neutral solid conducting sphere of radius $r_3=2\,m$ as shown in the figure. The outer surface is grounded using a resistance of 4Ω by closing key k. The heat dissipated on switching on the key is given by $\alpha\, mJ$. Fill the value of α.

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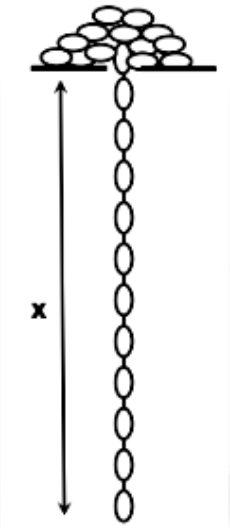


28. A satellite of mass m is in a circular orbit around an airless spherical planet of radius R. An asteroid of equal mass m falls radially towards the planet, starting at zero velocity from a very large distance. The satellite and the asteroid collide inelastically and stick together, moving in a new orbit such that it just misses the planet's surface. The radius of the satellite's original circular orbit is $\left(\alpha+\sqrt{11}\right)R$. Find the value of α.

29. A charged particle sits at the centre of a circular region containing a magnetic field, $\vec{B}$. The field $\vec{B}=\vec{B}(r)$ depends only on radial position r, and is perpendicular to the plane of circle. The total magnetic flux through the circle is zero. The particle is given a kick so that it leaves the circular region, then at the instant it leaves, its velocity makes an angle $n\times\frac{\pi}{4}$ rad with outward radial direction. Find n.

30. One end of the chain falls through a hole and pulls the remaining links after it in a steady flow. If the links which are initially at rest, acquire the velocity of the chain suddenly and without frictional resistance or interference from the support or from adjacent links. The acceleration of the chain when $x=\frac{L}{2}$ is $\frac{g}{N}$. Determine the value of N. (when $x=0$, then $v=0$, length of the chain is L and ρ is the mass per unit length of the chain, $g=$ acceleration due to gravity).

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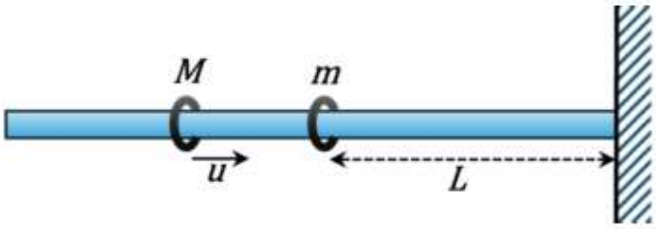


SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Two beads of mass M and m lie on a smooth horizontal rod whose one end is attached to a wall. Initially bead of mass m is at a distance $L = 3m$, from the wall and bead of mass M is given velocity $u = 2cm/s$, as shown in the figure. Consider all collisions to be elastic. Consider the size of the beads to be negligible. $\left(\frac{M}{m} = 10^4\right)$



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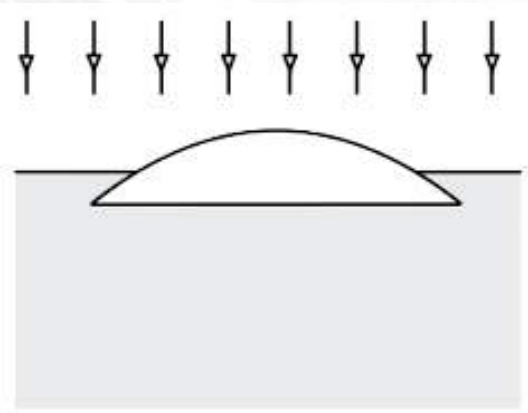
31. Determine the maximum speed of the bead (nearest integer in m/s) of mass m during the motion.

32. Determine the minimum distance (nearest integer in cm) of the bead of mass M from the wall during the motion.

PARAGRAPH-II

A thin plano-convex lens is partially immersed in water with its curved side up. Thickness of the lens in the middle is $4mm$. Aperture of the lens is horizontal as shown in the figure. Refractive index of water is $\frac{4}{3}$. If there is no reflection at any surface and no absorption in any medium, we obtain two equally bright images at depths $30cm$ and $40cm$.

(Here, equal brightness refer to equal energy of light used in image formation. Assume 100% refraction at all surfaces)



33. The depth of the planar surface of the lens from water surface in mm is

34. Radius of the curvature of the curved surface in cm is

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CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

35. A solid ionic compound consists of X^{2-} ions present at all face centres of a cubic unit cell. A^{2+} ions are present at all the corners and B^{4+} ions at the body centre .If the ions on a face of the unit cell are removed (disregarding change neutrality of the unit cell) the formula of the unit cell is

A) ABX_3 **B)** ABX_4 **C)** AB_2X_5 **D)** AB_2X_4

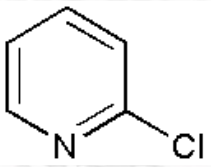
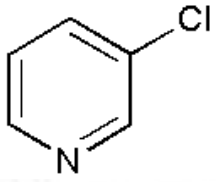
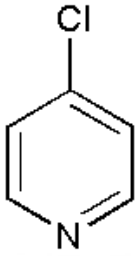
36. Which of the following lanthanide ion have the spin magnetic moment equal to sum of orbital and spin magnetic moment.

A) Ce^{3+} (A.No.58) **B)** Pr^{3+} (A.No.59) **C)** Gd^{3+} (A.No.64) **D)** Sm^{3+} (A.No.62)

37. Which of the following pair of ions are same colour in aq solution is

A) Cr^{2+}, Cu^{2+} **B)** V^{2+}, Cr^{3+} **C)** Fe^{2+}, Ni^{2+} **D)** All of these

38. Among the following, which one has no reaction (or) less reactive with RNH_2 ?

A)  **B)**  **C)**  **D)** All are equally reactive

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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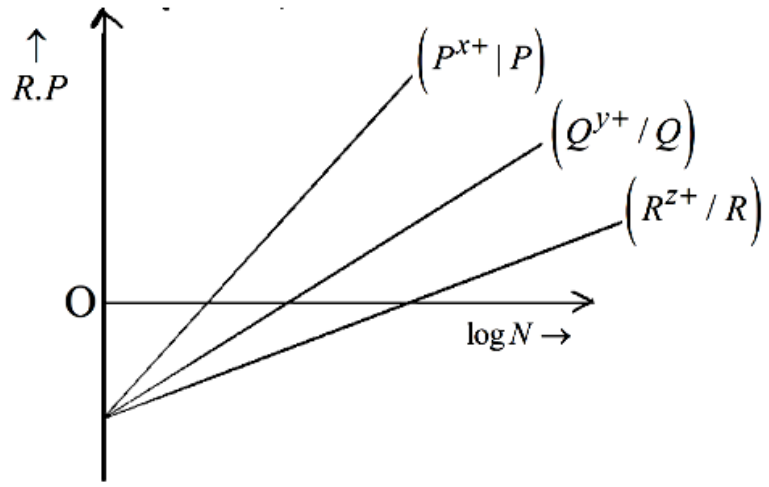
39. The reduction potential $(R.P)$ vs $\log N[N = \text{Normality of the oxidized form}]$ in the solution is shown below for three metal-metal ion half cells. The cell reactions involved are

$$P^{x+} + xe^- \rightleftharpoons P$$

$$Q^{y+} + ye^- \rightleftharpoons Q$$

$$R^{z+} + ze^- \rightleftharpoons R$$

The corresponding plots are shown below.



Which of the following statements **have to be incorrect** if it is known that $x \neq y \neq z$.

A) $E^{\circ}_{P^{x+}|P} = E^{\circ}_{Q^{y+}|Q} = E^{\circ}_{R^{z+}|R}$

B) $x > y > z$

C) $E^{\circ}_{P^{x+}|P} < E^{\circ}_{Q^{y+}|Q} > E^{\circ}_{R^{z+}|R}$

D) Some of these metals may be unable to displace H_2 from 1M HCl solution.

40. Which of the following statement(s) is/are correct?

A) Conc. nitric acid oxidizes arsenic to arsenic acid (H_3AsO_4)

B) Conc. nitric acid oxidizes antimony to Sb_2O_5

C) Conc. nitric acid oxidizes Bismuth to Bi^{3+}

D) Conc. nitric acid oxidizes tellurium to telluric acid (H_2TeO_4)

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41. $HC \equiv CH \xrightarrow{CuCl+NH_4Cl} P$ (Compound of 4 carbons)

$$P \xrightarrow[\text{Catalyst } CuCl \text{ and } NH_4Cl]{\text{Conc. } HCl} \underset{\text{Intermediate}}{[Q]} \xrightarrow{\text{Re arrangement}} R \xrightarrow[\text{Polymerisation}]{\text{Peroxide}} S$$

$$S \xrightarrow[\text{hydrogenation}]{\text{complete}} T$$

Among the following correct statement(s) is/are

A) Intermediate 'Q' formed as a result of $1,2-$addition of HCl to P.

B) 'S' is a synthetic rubber prepared by step growth polymerization.

C) 'T' is the homo polymer of $2-$chloro$-2-$butene

D) 'P' gives red precipitate with $\left(CuCl+NH_4OH\right)$

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

42. 0.02M solution of MCl_2 is titrated with 0.2M Na_4Z solution. The reaction proceed with formation of a complex between M^{2+} and Z^{4-}.$\left[M^{2+}\right]$ is measured at certain stages of titration (Both MCl_2 and Na_4Z are soluble ionic salts).

Vol of MX_2 solution	Vol of Na_4Z solution	$\left[M^{2+}\right]$
10mL 10 mL	0.995 mL 1.005 Ml	C_1 C_2

If $\frac{C_1}{C_2}=100$, the complex formation constant for the complex $\left[MZ\right]^{2-}$ is nearly $a\times10^b$, where a and b are both positive single digit integers. Calculate $a+b$.

43. We may define '$n-$factor' for a species involved in a chemical reaction as the ratio of its gram-molecular mass to its gram-equivalent mass for the reaction.

In how many of the following reactions (unbalanced) do both underlined species have identical $n-$factor?

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I. $\underline{Ca(NO_3)_2} + \underline{H_3PO_4} \rightarrow CaHPO_4 + HNO_3$

II. $\underline{HCHO} + K_2Cr_2O_7 + H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O + \underline{CO_2}$

III. $\underline{Cu_2S} + KMnO_4 + H_2SO_4 \rightarrow CuSO_4 + K_2SO_4 + MnSO_4 + H_2O + \underline{SO_2}$

IV. $KMnO_4 + \underline{HCl} \rightarrow KCl + MnCl_2 + H_2O + \underline{Cl_2}$

V. $KI + \underline{H_2O_2} \rightarrow KOH + \underline{I_2}$

VI. $\underline{CaCO_3} + HCl \rightarrow CaCl_2 + \underline{H_2O} + CO_2$

44. Consider the following compounds

1) XeF_n **2)** $\left(XeF_{n+1}^{+}\right)^{+}$ **3)** $\left(XeF_{n-1}^{-}\right)^{-}$ **4)** XeF_{n+2} **5)** XeO_3F_{n-2}

If value of n equals to number of unpaired electrons are present in $Cr_{(aq)}^{2+}$

Then calculate the $\frac{x}{y} =$

$x =$ Total number of covalent bond pairs in all the above species

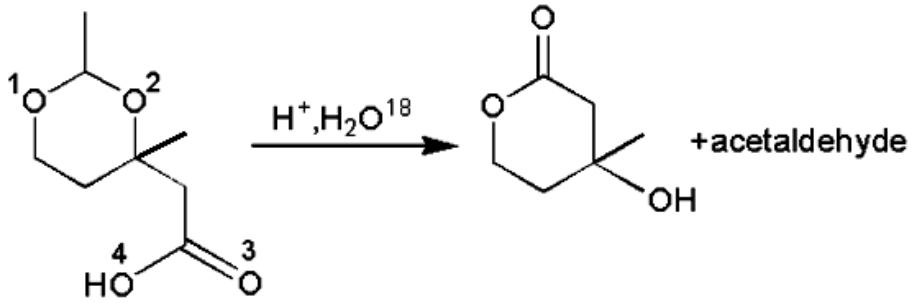
$y =$ Total number of lone pair on central atoms of all the above species

(round off nearest integer)

45. The no. of unpaired electrons in the e_g (axial) set of d-orbitals in the complex

$\left[Rh(H_2O)_3F_3\right]$ (A. No $= 45$)

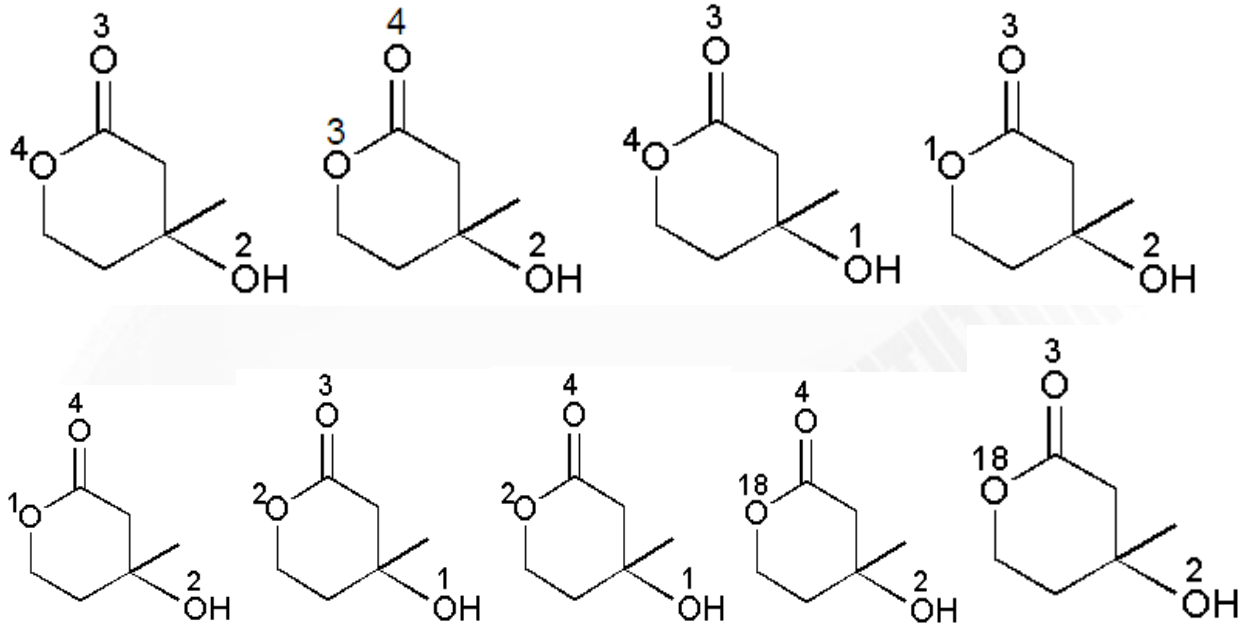
46.



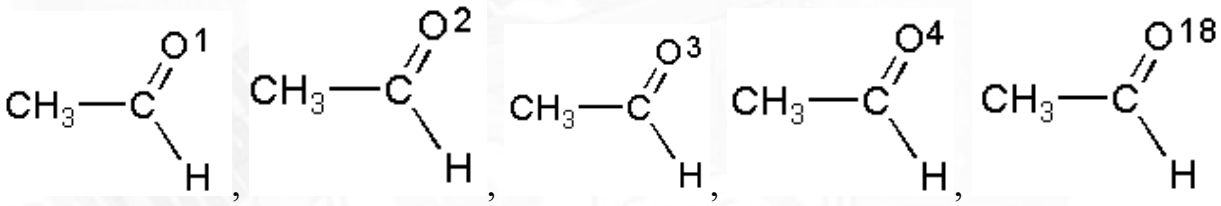
In the above reaction, four oxygens in the reactant are labeled as 1,2,3 and 4.

I) From the following list of structures, 'X' number of structures are possible for the product lactone derivative.

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II) From the following structures, 'Y' number of structures possible for the product acetaldehyde.



Sum of X and Y is

47. $$\underset{(I)}{\text{Aniline}} \xrightarrow{\text{Conc.}H_2SO_4} (II) \xrightarrow{200^\circ C} (III) \xrightarrow{+NH_3,\Delta} (IV)$$

$$C_6H_5NO_2 \xrightarrow[\substack{\text{Mononitration}\\ \text{II) excess } H_2/Ni}]{\text{I) Conc.}HNO_3+\text{Conc.}H_2SO_4,\Delta} (V) \xrightarrow[pH=4-5]{\text{diazonium salt of }(IV)} \text{Prontosil}$$

The antibiotic prontosil, in human body converts to the real active compound 'X'.'X' is one of the compounds I, II, III, IV, V involved in the above reaction sequence. Sum of the degree of unsaturation of prontosil and the number assigned to 'X' is

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SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

The chamber A shown below initially contains 0.04 moles of $XY(s)$ and 1 mole $Ne(g)$ at 300K. Chamber B contains $Ne(g)$ at 300K and 25 bar pressure. The surrounding are at a pressure of 25 bar. The cavity of chamber A is cubical with $H=10cm.$ The cavity of chamber B is cuboidal with $h_1=12cm, h_2=\frac{3}{2}cm$ and base area $=\ell^2=\frac{200}{3}cm^2.$ (All walls are rigid). $XY(s)$ is known known to dissociate into $X(g)$ and $Y(g)$ at temperatures above 680K according to the reaction

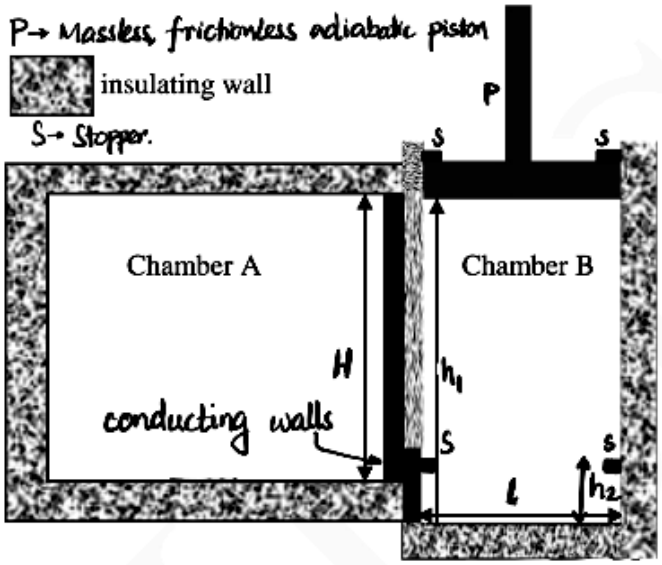
$$XY(s)\rightleftharpoons X(g)+Y(g) \qquad K_p=16\,\text{bar}^2$$

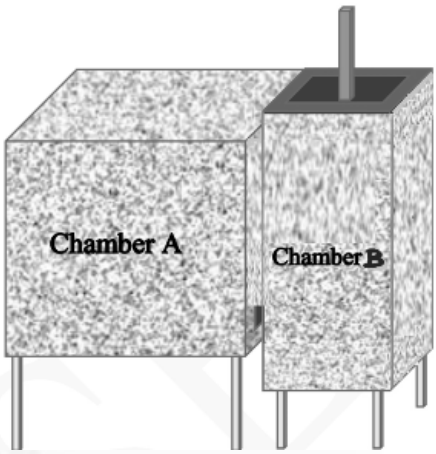
The internal energy change, $\Delta U_r^\circ=0$ for the dissociation and can be considered to be constant at all temperatures.

The gas in chamber B is compressed reversibly, till the piston reaches the lower stoppers. Assume that the compression happens adiabatically and any heat transfer between chambers A and B takes place after complete compression has been carried out. The piston is maintained at this position and equilibrium is allowed to be reached between chambers A and B.

$$\left[\begin{array}{l} Take\,C_{V_M}(Ne)=\frac{3R}{2};R=\frac{1}{12}\text{L bar mol}^{-1}\text{K}^{-1}=2\,\text{Cal mol}^{-1}\text{K}^{-1} \\ \ell n2=0.7,\ell n3=1.1,\ell n7=2.\text{Neglect the contribution of } XY,X,Y \\ \text{and walls of the chambers towards the heat capacities.} \end{array}\right]$$

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48. If the overall magnitude of entropy change of gas in chamber B is $x\,\text{cal}\,\text{K}^{-1}$, calculate $5x$.

49. What is the final pressure in chamber A (in bar)?

PARAGRAPH-II

Bond energy of some bonds given below

Bond	Bond energy (in K.Cal/mol)
$Cl-Cl$	84
$H-Cl$	103
$C-H$	101
$C-Cl$	85
$C-C$	83
$C=C$	146

Halogentation of alkanes is free radical substitution mechanism

$$CH_3-CH_3(g)+Cl_2(g)\xrightarrow{h\upsilon}CH_3CH_2Cl(g)+HCl(g)$$

Answer the following two questions regarding chlorination ethane by using above data.

50. Activation energy of chain initiation step is P K.Cal/mol, standard enthalpy of reaction of rate determining step is Q K.Cal/mol. Sum of P and Q is

51. In chain termination step, following reaction possible

I) Disproportionation of free radicals

II) Coupling of alkyl free radicals

III) Coupling of alkyl and halogen free radicals and

IV) Coupling of halogen free radicals.

Sum of standard enthalpy of reaction of reaction I and activation energies of reaction II, III, IV is (in K.Cal/mol).

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BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 10 ALL INDIA CATEGORY RANKS COUNT	BELOW 100 ALL INDIA CATEGORY RANKS COUNT	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT
31	95	10	98	579

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2025 22,094

*DLP/AITS

JEE 2025 STARS SHINE BRIGHT

Sri Chaitanya Tops JEE ADVANCED

ALL INDIA OPEN CATEGORY RANKS



RUTVIK SAI
H.T.No. 256055278 (OBC-NCL)



MAJID MUJAHID HUSAIN
H.T.No. 251134112*



UJJWAL KESARI
H.T.No. 252016104*



AKSHAT KUMAR CHAURASIA
H.T.No. 254065055*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	NUMBER OF QUALIFIED RANKS
29	113	205	745	4,212

*DLP/AITS

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